

Daily Number Theory Drill #5

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Bounding Diophantine Equations · Inequality Logic · Factor Searches.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Last digits, primes, and basic properties. Pure recognition.

- Find the last digit of 3^{2025} .
- Factorise 1001 completely into primes.
- Find $\gcd(42, 105)$.
- Find the number of positive divisors of 120.
- Evaluate $5^{10} \pmod{6}$.
- Find the smallest positive integer x such that $2x \equiv 1 \pmod{5}$.
- Find the smallest integer $n > 1$ such that $n \equiv 1 \pmod{3}$ and $n \equiv 1 \pmod{4}$.
- Write the prime factorisation of 360.
- Calculate $7^3 \pmod{10}$.
- Find the largest prime factor of 9999.

Section B Manipulation Drills

(≤50 s each)

Modular arithmetic, exponentiation, divisibility rules. Look for patterns.

- The greatest power of 7 which is a factor of $50!$ is 7^k . What is k ? (*after SMC 2023 Q12*)
- Find the smallest positive integer n such that $15n$ is a perfect cube.
- Find the remainder when $1! + 2! + 3! + \cdots + 100!$ is divided by 12.
- Which is larger, 3^{40} or 4^{30} ?

5. Find the smallest positive integer x such that $3x \equiv 4 \pmod{7}$.
6. Find the last two digits of 7^{2022} .
7. How many positive factors of 1000 are perfect squares?
8. Find all integers x such that $0 \leq x < 8$ and $x^2 \equiv 1 \pmod{8}$.
9. Given $2^x 3^y = 12^3$, find $x + y$.
10. Find all pairs of positive integers (x, y) such that $xy = x + y + 3$.

Section C Substitution & Structure

(≤90 s each)

Synoptic links: use algebraic identities and combinatorial counting to solve these number theory problems.

1. The expression $\frac{n^2+3n+5}{n+1}$ takes integer values for certain positive integer values of n . What is the sum of all such positive integer values of n ? (after SMC 2023 Q19)
2. How many of the numbers 6, 7, 8, 9, 10 are factors of the sum $2^{2024} + 2^{2023} + 2^{2022}$? (after SMC 2023 Q10)
3. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 17$.
4. How many ordered pairs of positive integers (a, b) satisfy $\text{lcm}(a, b) = 300$?
5. How many positive integers $n \leq 100$ have exactly 3 positive divisors?
6. Find the sum of the digits of the largest integer n such that $n^3 + 100$ is a multiple of $n + 10$.
7. Find all triples of positive integers (x, y, z) such that $x \leq y \leq z$ and $xyz = x + y + z$. (after SMC 2022 Q21)
8. Find the sum of all integer values of n for which $n^2 + 4n + 3$ is a perfect square. (after AI Prompts D5)

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find all prime numbers p such that $p^2 + 2$ is also a prime number.
2. Find all pairs of positive integers (x, y) such that $x^2 y = x^2 + 3y + 2$. (after BMO1 1995 Q1)
3. How many pairs of positive integers (x, y) with $x \leq y$ satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$? (after SMC 2021 Q24)
4. Find the remainder when $1^{2025} + 2^{2025} + \dots + 10^{2025}$ is divided by 11.

5. Find the sum of all integers n for which $n^4 + 4$ is a prime number.

“Number theory is the queen of mathematics.” — Carl Friedrich Gauss

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths